

Si Observables Results

Contents

1	Brief Note on Implementation	2
2	Convergence Tests	2
3	Silicon Equation of State	2
4	Elastic Properties of Silicon	3
5	Silicon Observables	4

1 Brief Note on Implementation

All calculations were performed using the Quantum ESPRESSO (QE) software package and executed on the ARCHER2 supercomputing cluster. Job submission was handled via SLURM scripts adapted from the standard templates, with consistent HPC parameters applied across all runs. Each job was run on a single exclusive node with 36 MPI tasks (`--tasks-per-node=36`) and no OpenMP threading (`OMP_NUM_THREADS=1`). K-point parallelisation was enabled using the `-npool` flag, with six pools (`-npool 6`) used to distribute k-points across tasks, thereby improving the efficiency of Brillouin zone integrations while maintaining balanced workload distribution.

2 Convergence Tests

Prior to the production calculations, convergence tests were performed with respect to the plane-wave cutoff energy and Brillouin zone sampling, using the total energy per atom as the reference metric. A plane-wave cutoff of 120 Ry was adopted, as this value ensures convergence of the total energy to within 5 meV relative to the highest cutoff accessible within the available computational resources. Similarly, a $10 \times 10 \times 10$ Monkhorst-Pack k -point mesh was selected, providing convergence of the total energy to better than 1 meV. These thresholds were chosen to guarantee numerical accuracy while maintaining computational efficiency and practical runtimes for the large number of calculations required in this work. For brevity, the detailed convergence curves are not shown here, but the selected parameters are sufficient to ensure the reliability of the results reported below.

3 Silicon Equation of State

The equation of state of silicon was studied by varying the input lattice parameter in regular intervals between 95% and 105% of its initial value. A total of 200 data points were obtained. The extracted energy values of a Si atom, (half that of the unit cell) were fitted to three known models for the equation of state, expressing energy as a function of the relative linear compression η . The fitting of the models to the data, as well as the calculation of the best-fit parameters and associated uncertainties, was performed using the tools integrated in the `scipy.optimize` library in Python. In this case, the values E_0 and V_0 were treated as fitting parameters, along with the bulk modulus and its derivative at zero applied pressure, B_0 and B'_0 .

The calculated data points together with the fitted EOS models are presented in Figure 1. For the fitting procedure, initial guesses for the equilibrium parameters were taken from literature values: $V_0 = 20.0 \text{ \AA}^3/\text{atom}$, $B_0 = 100 \text{ GPa}$, $B'_0 = 5$, and $E_0 = -130 \text{ eV}$. For clarity of presentation, only one third of the computed data points are shown in the figure; however, the full set of 200 points was used in all fittings and parameter estimations. The best-fit parameters and associated uncertainties for both cases can be found in Table 1.

EOS	$V_0 \text{ (\AA}^3\text{)}$	$E_0 \text{ (eV)}$	$B_0 \text{ (GPa)}$	B'_0
Murnaghan	$20.4513 \pm 4 \cdot 10^{-4}$	-130.56770 ± 10^{-5}	88.08 ± 0.03	4.124 ± 0.008
Birch-Murnaghan	$20.4476 \pm 2 \cdot 10^{-4}$	-130.56775 ± 10^{-5}	88.47 ± 0.01	4.239 ± 0.003
Vinet	$20.44590 \pm 4 \cdot 10^{-5}$	-130.567774 ± 10^{-6}	88.751 ± 0.003	$4.2937 \pm 9.0 \cdot 10^{-4}$

Table 1: Best-fit parameters and associated uncertainties for the three equations of state fitted to the data.

The quality of the fits was assessed through the coefficient of determination R^2 , which yielded values of 0.99333, 0.99386, and 0.99331 for the Murnaghan, third-order Birch-Murnaghan, and Vinet EOS, respectively. Although the Birch-Murnaghan form provides a slightly better description, the improvement is marginal. The best-fit parameters obtained from the three models are also consistent with one another. As shown in Table 1, the equilibrium volume varies only slightly between models, ranging from $V_0 = 20.4459 \text{ \AA}^3$ (Vinet) to $V_0 = 20.4513 \text{ \AA}^3$ (Murnaghan), representing an overestimation of about 2.2% relative to the experimental reference of $V_0 = 20.0 \text{ \AA}^3$. Such

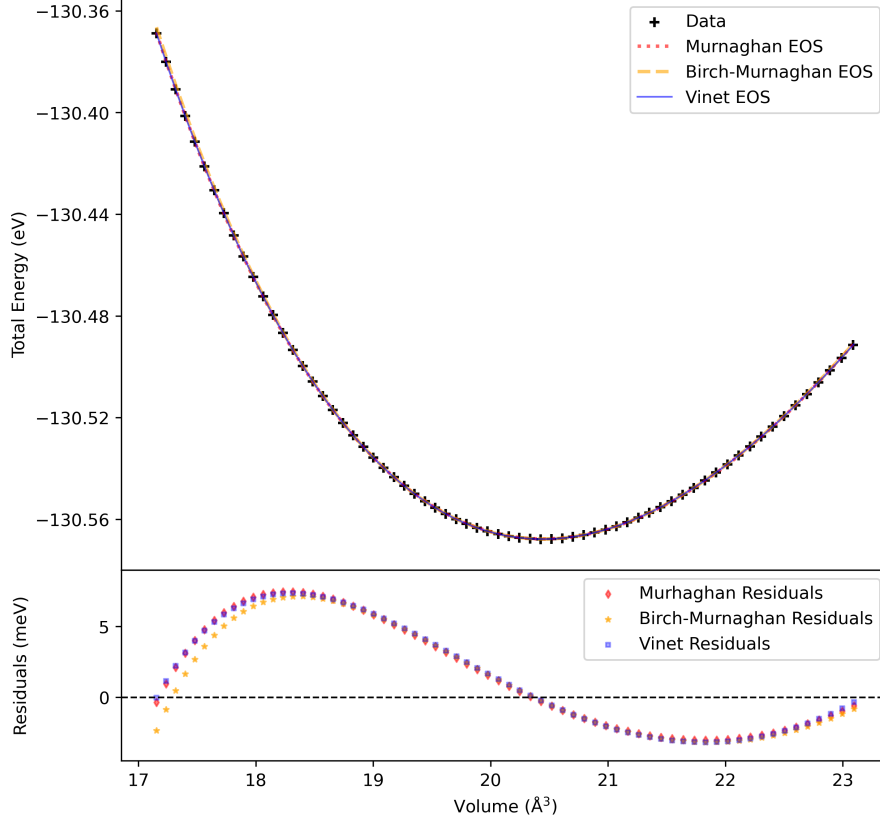


Figure 1: Calculated total energy as a function of volume per atom, together with fits to three standard equations of state for the $E(V)$ relation. The main vertical axis is given in eV, while the residuals are plotted on a secondary axis in meV.

overestimation of V_0 is a well-known feature of GGA functionals, including PBE, and naturally leads to an underestimation of the bulk modulus since

$$B_0 = V_0 \left(\frac{\partial^2 E}{\partial V^2} \right)_{V=V_0} \Rightarrow B_0 \propto \frac{1}{V_0}. \quad (1)$$

From this relation, the observed volume overestimation would suggest $B_0 \approx 97$ GPa, whereas the fits yield smaller values between 88.08 GPa (Murnaghan) and 88.751 GPa (Vinet). The residual discrepancy is likely due to limitations of the pseudopotential, which may underestimate the converged energies. The pressure derivative B'_0 shows only minor variation between models, from 4.124 (Murnaghan) to 4.2937 (Vinet), confirming the overall robustness of the results. Finally, it was verified that the fitted parameters are not strongly affected by the subset of data points considered.

4 Elastic Properties of Silicon

The elastic coefficients c_{11} and c_{12} were calculated from stress-strain relations obtained within DFT. The equilibrium lattice constant, $a_0 = 10.3345$ Å, as determined from the Vinet equation of state, was used as the reference structure. At this value, the residual stress tensor components are -2.3×10^{-7} Ry/bohr³ (≈ -0.03 kbar), which is negligible for practical purposes. A $10 \times 10 \times 10$ Monkhorst-Pack k -point grid and a plane-wave cutoff of 120 Ry were employed, yielding total energies converged to within 1 meV/atom. Although no separate convergence study was carried out specifically for the elastic constants, these parameters are sufficiently accurate for extracting reliable stress-strain relations.

The coefficients c_{11} and c_{12} were obtained by fitting a linear function to the calculated stress-strain data. Stress tensor components were evaluated with the input flags `tstress = .true.` and

`tpnfor = .true.` enabled in QE. A total of 90 data points were generated by applying uniaxial strains in the range $-0.01 \leq \epsilon \leq 0.01$ to the tetragonal cell. The distortions were introduced by varying the c/a ratio by up to $\pm 1\%$ relative to the equilibrium cubic value of 1, thereby probing the anisotropic response required to extract the independent elastic constants. In this setup, the imposed variation of c/a corresponds to introducing a strain along the z -axis.

The data points and lines of best fit are shown in Figure 1.

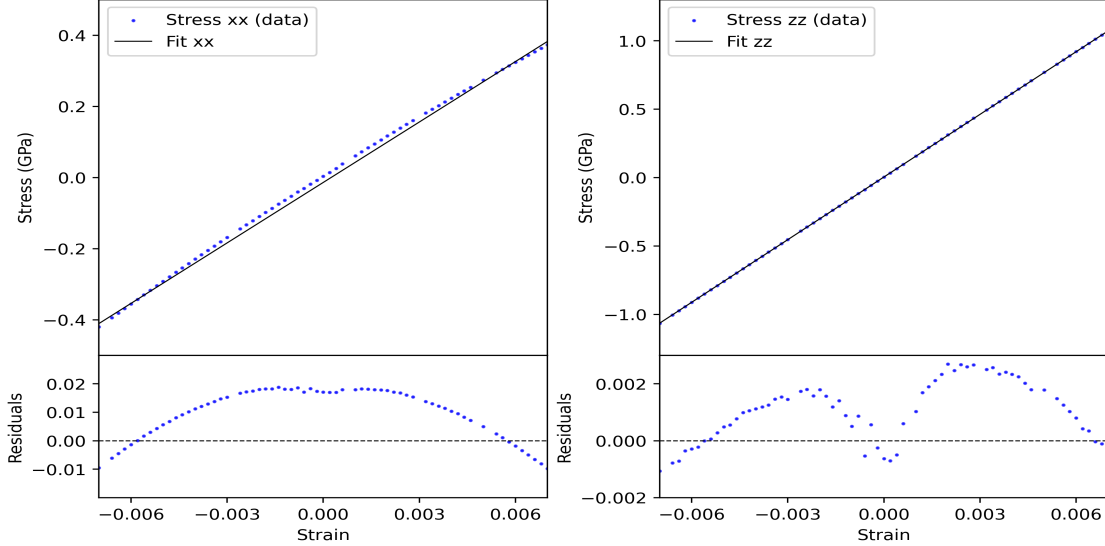


Figure 2: Graph showing components of the stress tensor the σ_{11} and σ_{33} against the strain rate ϵ_{33} . The lines are those of best fit to the data. **Left Panel:** Normal Stress along x-axis (σ_{11}) against normal strain along z-axis (ϵ_{33}) **Right Panel:** Normal Stress along z-axis (σ_{33}) against normal strain along z-axis (ϵ_{33}). A residual plot has been added to show the deviation from the linear trend.

The best-fit parameters were obtained using the `curve_fit` function from the `scipy.optimize` library. The quality of the fits was assessed via the coefficient of determination (R^2), yielding $c_{12} = 56.7 \pm 0.3$ GPa with $R^2 = 0.997$ and $c_{11} = 152.65 \pm 0.04$ GPa with $R^2 = 0.999$. These values deviate from the experimental references of 64 GPa and 166 GPa by approximately 12% and 8%, respectively. Nevertheless, they are consistent with values reported in the literature for first-principles calculations employing the PBE functional, which typically yield $c_{11} \approx 157.8$ GPa and $c_{12} \approx 55.7$ GPa.

In the small-strain regime, the stress-strain relation is expected to be strictly linear by definition of the elastic constants. The systematic curvature observed in the residuals, particularly in the left-hand panel (c_{12}), indicates that the data may have been affected by numerical noise or incomplete convergence in the underlying DFT calculations. In the case of c_{11} (right-hand panel), the deviations are less pronounced but still display a weak structured pattern, which could likewise originate from numerical artefacts. Overall, while the linear model remains the theoretically correct description in this regime, the residual features highlight the sensitivity of stress-based fitting to small numerical inaccuracies.

5 Silicon Observables

Several macroscopic observables of silicon were derived from the calculated equilibrium volume V_0 and the single-crystal elastic constant $c_{11} = 152.65 \pm 0.04$ GPa. In particular, the longitudinal sound speed along the $[100]$ direction was obtained from the relation

$$c_L([100]) = \sqrt{\frac{c_{11}}{\rho}},$$

where the density ρ was determined using V_0 from the EOS fits. From these values, the density, atomic number density, longitudinal wave velocity, and Debye temperature were computed, with uncertainties propagated consistently from the EOS fits and the elastic constant. The results are summarised in Table 2.

EOS	Density (g/cm ³)	c_L ([100]) (m/s)	n (atoms/Å ³)	Θ_D (K)
Murnaghan	$2.28039 \pm 4 \cdot 10^{-5}$	8181.71 ± 1.07	$0.048910 \pm 1.0 \cdot 10^{-6}$	890.81 ± 0.12
Birch–Murnaghan	$2.28081 \pm 2 \cdot 10^{-5}$	8180.95 ± 1.07	$0.0489053 \pm 5.0 \cdot 10^{-7}$	890.70 ± 0.12
Vinet	$2.281000 \pm 4 \cdot 10^{-6}$	8180.61 ± 1.07	$0.0489096 \pm 1.0 \cdot 10^{-7}$	890.69 ± 0.12

Table 2: Density, longitudinal [100] sound velocity from c_{11} , atomic number density, and Debye temperature (computed using c_L only). Uncertainties are obtained by propagating the EOS-fit errors and the c_{11} error.

The calculated longitudinal sound velocity ($c_L^{[100]} \approx 8180$ m/s) is lower than the experimental range of 8430–8500 m/s at ambient conditions [?]. This discrepancy arises primarily from the reduced value of the elastic constant c_{11} obtained in our calculations (152.65 GPa), compared to the experimental value of about 167 GPa. Because the sound velocity scales as $c_L^{[100]} \propto \sqrt{c_{11}/\rho}$, even a modest reduction in c_{11} directly propagates into a noticeable decrease in c_L . This highlights the strong sensitivity of elastic wave velocities to the accuracy of the underlying elastic constants.

The computed densities are slightly lower than the experimental value of ~ 2.33 g/cm³, reflecting the overestimation of the equilibrium volume V_0 that is characteristic of GGA functionals. In contrast, the atomic number densities agree almost exactly with the commonly reported value of 0.049 atoms/Å³, with deviations below 0.01%, indicating that the structural description is well captured. The Debye temperatures, however, are substantially higher than the experimental value of ~ 645 K. This overestimation is expected because the present calculation uses only the longitudinal sound speed c_L , whereas a realistic estimate should be based on the effective sound velocity, c_s , defined by $3/c_s^3 = 1/c_L^3 + 2/c_T^3$, or on an orientational average of the single-crystal velocities.